Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: 6-X #2

Physical address of dairy:

6960 Crane RD	Oakdale	Stanislaus	95361
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1956

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X025-X008-XXXX

B. OPERATORS

Osmundson, Gary

Operator name: Osmundson, Gary

Telephone no.: (209) 845-2114 (Landline) (Cellular)

P.O. Box 12290	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Vierra Brothers Dairy

Legal owner name: Vierra Brothers Dairy

Telephone no.: (209) 847-5705 (Landline) (Cellular)

6960 Crane RD	Oakdale	CA	95361
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	460	65	335	180	350	0
Number under roof	0	0	0	0	0	0
Maximum number	500	70	335	180	350	0
Average number	460	65	335	180	350	0
Avg live weight (lbs)	1,300	1,350	950	700		

Predominant milk cow breed: Other

Average milk production: 76 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,413.96 *tons per reporting period*

Total nitrogen from manure: 237,130.96 *lbs per reporting period*

After ammonia losses (30% loss applied): 165,991.67 *lbs per reporting period*

Total phosphorus from manure: 37,998.00 *lbs per reporting period*

Total potassium from manure: 89,907.56 *lbs per reporting period*

Total salt from manure: 231,537.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,687,250 *gallons*

Total nitrogen generated: 32,494.95 *lbs*

Total phosphorus generated: 1,797.34 *lbs*

Total potassium generated: 31,581.95 *lbs*

Total salt generated: 185,177.55 *lbs*

	3,687,250	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,687,250	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal OID	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	Compost	8,970.00 *ton*	Dry-weight	36.7		13,600.00	5,360.00	22,500.00		69.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	154,441.87	60,868.27	255,510.45	7,869,721.86
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	154,441.87	60,868.27	255,510.45	7,869,721.86

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1 B	8	8	2	process wastewater	X063-X025-X008-XXXX
2 A	34	34	1	none	X063-X025-X008-XXXX
3 E	21	21	1	none	X063-X025-X008-XXXX
4 F	30	30	2	both	X063-X027-X002-XXXX
5 G	47	47	2	both	X063-X027-X002-XXXX
6 H	20	20	2	both	X063-X027-X002-XXXX
7 I	14	14	2	both	X063-X027-X002-XXXX
8 J	40	40	2	manure	X063-X027-X024-XXXX X063-X027-X052-XXXX
Totals for areas that were used for application	159	159	12		
Totals for areas that were not used for application	55	55	2		
Land application area totals	214	214	14		

B. CROPS AND HARVESTS

1 B

Field name: 1 B

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	126.40 *ton*	As-is		70.2	3,970.00	1,170.00	11,900.00		8.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	125.45	36.97	376.04	836.21

1 B

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	202.00 *ton*	As-is		63.6	5,100.00	720.00	7,280.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	25.25	257.55	36.36	367.64	1,176.45

2 A

Field name: 2 A

10/31/2018: Almond Field Run

Crop: Almond Field Run Acres planted: 34 Plant date: 10/31/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	221.40 *ton*	As-is		11.6	6,630.00	1,120.00	41,300.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	6.51	86.35	14.59	537.87	1,473.64

3 E

Field name: 3 E

3 E

01/06/2017: Almond Field Run

Crop: Almond Field Run Acres planted: 21 Plant date: 01/06/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	140.70 *ton*	As-is		11.6	6,630.00	1,120.00	41,300.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	6.70	88.84	15.01	553.42	1,516.24

4 F

Field name: 4 F

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	474.00 *ton*	As-is		68.0	3,380.00	1,000.00	11,000.00		7.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	106.81	31.60	347.60	797.84

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	962.00 *ton*	As-is		72.2	3,480.00	1,010.00	8,740.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	32.07	223.18	64.77	560.53	927.11

5 G

Field name: 5 G

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	742.60 *ton*	As-is		68.0	3,380.00	1,000.00	11,000.00		7.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	106.81	31.60	347.60	797.84

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,450.00 *ton*	As-is		73.6	3,930.00	520.00	6,910.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.85	242.49	32.09	426.36	863.34

6 H

Field name: 6 H

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	316.00 *ton*	As-is		68.0	3,380.00	1,000.00	11,000.00		7.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	106.81	31.60	347.60	797.84

6 H

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	625.00 *ton*	As-is		74.0	3,950.00	540.00	5,890.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.25	246.88	33.75	368.13	991.25

7 I

Field name: 7 I

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	173.80 *ton*	As-is		68.0	3,380.00	1,000.00	11,000.00		7.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	106.81	31.60	347.60	797.84

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	371.00 *ton*	As-is		74.0	3,220.00	690.00	5,730.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.73	217.20	46.54	386.51	999.68

8 J

Field name: 8 J

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	426.60 *ton*	As-is		70.2	3,970.00	1,170.00	11,900.00		8.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	125.45	36.97	376.04	836.21

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	840.24 *ton*	As-is		72.4	4,530.00	480.00	5,740.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.12	281.95	29.88	357.26	1,305.55

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 B - 10/20/2022: Oats, silage-soft dough

Field name: 1 B

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.28	14.75	139.08	1,189.66	162,000.00 *gal*
Application event totals		93.28	14.75	139.08	1,189.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.96	12.25	149.75	889.71	175,500.00 *gal*
Application event totals		60.96	12.25	149.75	889.71	

1 B - 06/10/2023: Corn, silage

Field name: 1 B

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	174.41	5.26	93.55	282.23	76,000.00 *gal*
Application event totals		174.41	5.26	93.55	282.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

1 B - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.14	0.00	0.00	58.58	1,080,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	54.92	1,012,500.00 *gal*
Application event totals		1.07	0.00	0.00	54.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	54.92	1,012,500.00 *gal*
Application event totals		1.07	0.00	0.00	54.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.36	8.75	165.94	1,223.46	162,000.00 *gal*
Canal OID	Ground water	0.85	0.00	0.00	43.94	810,000.00 *gal*
Application event totals		100.22	8.75	165.94	1,267.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.00	0.00	0.00	51.26	945,000.00 *gal*
Application event totals		1.00	0.00	0.00	51.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.00	0.00	0.00	51.26	945,000.00 *gal*
Application event totals		1.00	0.00	0.00	51.26	

1 B - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.00	0.00	0.00	51.26	945,000.00 *gal*
Application event totals		1.00	0.00	0.00	51.26	

2 A - 10/31/2018: Almond Field Run

Field name: 2 A

Crop: Almond Field Run

Plant date: 10/31/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	34.00	0.00	0.00	0.00	
Application event totals		34.00	0.00	0.00	0.00	

09/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.00	0.00	0.00	0.00	3,406,800.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	3,349,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	3,349,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

2 A - 10/31/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	3,349,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	3,349,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

05/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

2 A - 10/31/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.00	0.00	0.00	0.00	3,406,800.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

2 A - 10/31/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	3,391,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

11/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	30.60	0.00	0.00	0.00	
Application event totals		30.60	0.00	0.00	0.00	

3 E - 01/06/2017: Almond Field Run

Field name: 3 E

Crop: Almond Field Run

Plant date: 01/06/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	34.00	0.00	0.00	0.00	
Application event totals		34.00	0.00	0.00	0.00	

09/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.00	0.00	0.00	0.00	2,104,200.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.00	0.00	0.00	0.00	2,104,200.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

3 E - 01/06/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	2,068,500.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	2,068,500.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	2,068,500.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.83	0.00	0.00	42.74	2,068,500.00 *gal*
Application event totals		0.83	0.00	0.00	42.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

3 E - 01/06/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

3 E - 01/06/2017: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.84	0.00	0.00	43.29	2,094,750.00 *gal*
Application event totals		0.84	0.00	0.00	43.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	30.60	0.00	0.00	0.00	
Application event totals		30.60	0.00	0.00	0.00	

4 F - 10/20/2022: Oats, silage-soft dough

Field name: 4 F

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	143.47	45.06	251.47	3,538.34	300.00 *ton*
Application event totals		143.47	45.06	251.47	3,538.34	

4 F - 06/10/2023: Corn, silage

Field name: 4 F

Crop: Corn, silage

Plant date: 06/10/2023

4 F - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	174.41	5.26	93.55	282.23	285,000.00 *gal*
Application event totals		174.41	5.26	93.55	282.23	

06/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.10	0.00	0.00	56.63	3,915,000.00 *gal*
Application event totals		1.10	0.00	0.00	56.63	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.06	0.00	0.00	54.68	3,780,000.00 *gal*
Application event totals		1.06	0.00	0.00	54.68	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.06	0.00	0.00	54.68	3,780,000.00 *gal*
Application event totals		1.06	0.00	0.00	54.68	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.06	0.00	0.00	54.68	3,780,000.00 *gal*
Application event totals		1.06	0.00	0.00	54.68	

4 F - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.74	8.17	154.88	1,141.90	567,000.00 *gal*
Canal OID	Ground water	0.80	0.00	0.00	41.01	2,835,000.00 *gal*
Application event totals		93.54	8.17	154.88	1,182.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.02	0.00	0.00	52.72	3,645,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.99	0.00	0.00	50.77	3,510,000.00 *gal*
Application event totals		0.99	0.00	0.00	50.77	

5 G - 10/20/2022: Oats, silage-soft dough

Field name: 5 G

Crop: Oats, silage-soft dough Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	143.47	45.06	251.47	3,538.34	470.00 *ton*
Application event totals		143.47	45.06	251.47	3,538.34	

5 G - 06/11/2023: Corn, silage

Field name: 5 G

Crop: Corn, silage Plant date: 06/11/2023

5 G - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	174.41	5.26	93.55	282.23	446,500.00 *gal*
Application event totals		174.41	5.26	93.55	282.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.11	0.00	0.00	57.34	6,210,000.00 *gal*
Application event totals		1.11	0.00	0.00	57.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	54.84	5,940,000.00 *gal*
Application event totals		1.07	0.00	0.00	54.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.84	8.44	160.06	1,180.08	918,000.00 *gal*
Canal OID	Ground water	0.82	0.00	0.00	42.38	4,590,000.00 *gal*
Application event totals		96.66	8.44	160.06	1,222.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.07	0.00	0.00	54.84	5,940,000.00 *gal*
Application event totals		1.07	0.00	0.00	54.84	

5 G - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.04	0.00	0.00	53.60	5,805,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.04	0.00	0.00	53.60	5,805,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.97	0.00	0.00	49.86	5,400,000.00 *gal*
Application event totals		0.97	0.00	0.00	49.86	

6 H - 10/20/2022: Oats, silage-soft dough

Field name: 6 H

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	143.47	45.06	251.47	3,538.34	200.00 *ton*
Application event totals		143.47	45.06	251.47	3,538.34	

6 H - 06/13/2023: Corn, silage

Field name: 6 H

Crop: Corn, silage

Plant date: 06/13/2023

6 H - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	174.41	5.26	93.55	282.23	190,000.00 *gal*
Application event totals		174.41	5.26	93.55	282.23	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.14	0.00	0.00	58.58	2,700,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

06/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.08	0.00	0.00	55.65	2,565,000.00 *gal*
Application event totals		1.08	0.00	0.00	55.65	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.08	0.00	0.00	55.65	2,565,000.00 *gal*
Application event totals		1.08	0.00	0.00	55.65	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.05	8.46	160.41	1,182.68	391,500.00 *gal*
Canal OID	Ground water	0.82	0.00	0.00	42.47	1,957,500.00 *gal*
Application event totals		96.88	8.46	160.41	1,225.15	

6 H - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.02	0.00	0.00	52.72	2,430,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.02	0.00	0.00	52.72	2,430,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.97	0.00	0.00	49.79	2,295,000.00 *gal*
Application event totals		0.97	0.00	0.00	49.79	

7 I - 10/20/2022: Oats, silage-soft dough

Field name: 7 I

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	143.47	45.06	251.47	3,538.34	110.00 *ton*
Application event totals		143.47	45.06	251.47	3,538.34	

7 I - 06/14/2023: Corn, silage

Field name: 7 I

Crop: Corn, silage

Plant date: 06/14/2023

7 I - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	174.41	5.26	93.55	282.23	104,500.00 *gal*
Application event totals		174.41	5.26	93.55	282.23	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.14	0.00	0.00	58.58	1,485,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	53.26	1,350,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.26	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	53.26	1,350,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.26	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.03	0.00	0.00	53.26	1,350,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.26	

7 I - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.34	8.22	155.89	1,149.31	209,250.00 *gal*
Canal OID	Ground water	0.80	0.00	0.00	41.27	1,046,250.00 *gal*
Application event totals		94.14	8.22	155.89	1,190.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.98	0.00	0.00	50.59	1,282,500.00 *gal*
Application event totals		0.98	0.00	0.00	50.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.98	0.00	0.00	50.59	1,282,500.00 *gal*
Application event totals		0.98	0.00	0.00	50.59	

8 J - 10/20/2022: Oats, silage-soft dough

Field name: 8 J

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	143.47	45.06	251.47	3,538.34	270.00 *ton*
Application event totals		143.47	45.06	251.47	3,538.34	

8 J - 06/26/2023: Corn, silage

Field name: 8 J

Crop: Corn, silage

Plant date: 06/26/2023

8 J - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	259.31	82.14	152.14	12,784.00	400.00 *ton*
Application event totals		259.31	82.14	152.14	12,784.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.14	0.00	0.00	58.58	3,645,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.05	0.00	0.00	54.24	3,375,000.00 *gal*
Application event totals		1.05	0.00	0.00	54.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.05	0.00	0.00	54.24	3,375,000.00 *gal*
Application event totals		1.05	0.00	0.00	54.24	

8 J - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.05	0.00	0.00	54.24	3,375,000.00 *gal*
Application event totals		1.05	0.00	0.00	54.24	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	3,240,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	1.01	0.00	0.00	52.07	3,240,000.00 *gal*
Application event totals		1.01	0.00	0.00	52.07	

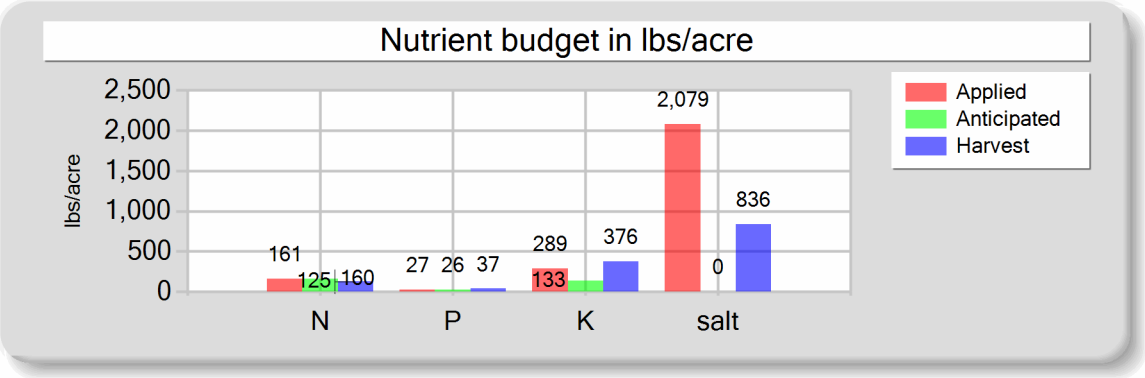
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal OID	Ground water	0.93	0.00	0.00	47.73	2,970,000.00 *gal*
Application event totals		0.93	0.00	0.00	47.73	

B. NUTRIENT BUDGET

1 B - 10/20/2022: Oats, silage-soft dough

Field name: 1 B Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.24	27.00	288.83	2,079.38
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	161.24	27.00	288.83	2,079.38
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.45	36.97	376.04	836.21
Nutrient balance	35.79	-9.97	-87.21	1,243.16
Applied to removed ratio	1.29	0.73	0.77	2.49

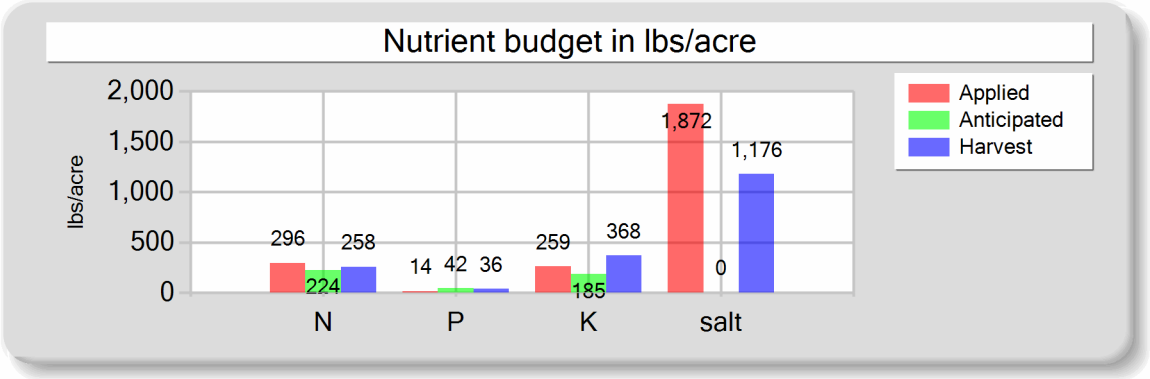
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
337,500.00 *gallons*
12.43 *acre-inches*
1.55 *inches/acre*

Total harvests for the crop
1 *harvests*

1 B - 06/10/2023: Corn, silage

Field name: 1 B Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	273.77	14.01	259.49	1,505.69
Fresh water	7.11	0.00	0.00	366.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.89	14.01	259.49	1,871.83
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	257.55	36.36	367.64	1,176.45
Nutrient balance	38.34	-22.35	-108.15	695.38
Applied to removed ratio	1.15	0.39	0.71	1.59

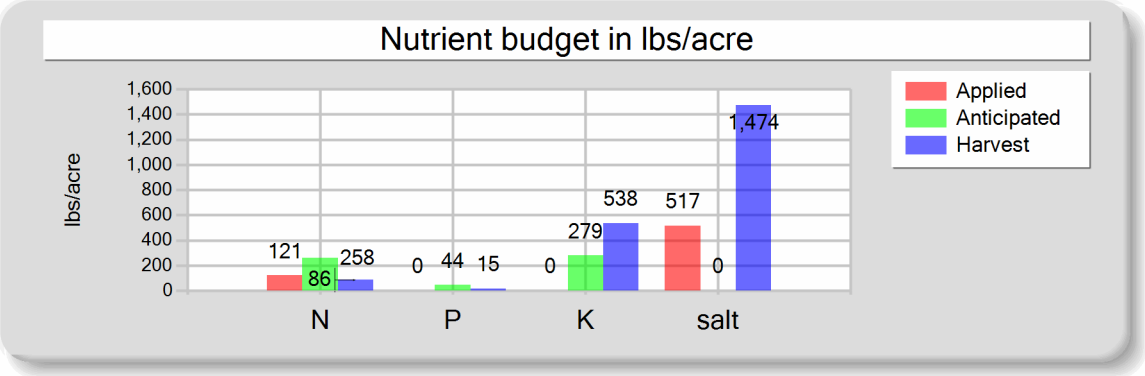
Fresh water applied	
6,750,000.00	*gallons*
248.58	*acre-inches*
31.07	*inches/acre*

Process wastewater applied	
238,000.00	*gallons*
8.76	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 A - 10/31/2018: Almond Field Run

Field name: 2 A Crop: Almond Field Run Plant date: 10/31/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	96.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.05	0.00	0.00	517.26
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	120.65	0.00	0.00	517.26
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	86.35	14.59	537.87	1,473.64
Nutrient balance	34.30	-14.59	-537.87	-956.38
Applied to removed ratio	1.40	0.00	0.00	0.35

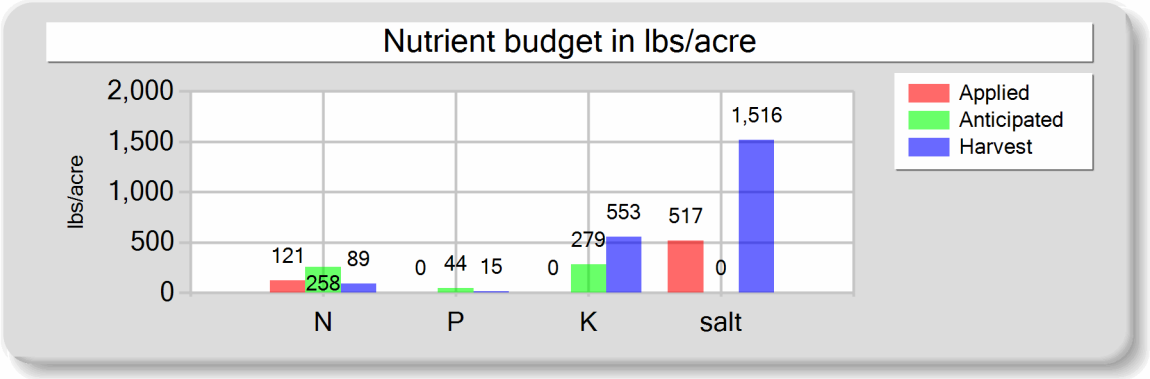
Fresh water applied	
47,341,600.00	*gallons*
1,743.43	*acre-inches*
51.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 E - 01/06/2017: Almond Field Run

Field name: 3 E Crop: Almond Field Run Plant date: 01/06/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	96.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.05	0.00	0.00	517.26
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	120.65	0.00	0.00	517.26
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	88.84	15.01	553.42	1,516.24
Nutrient balance	31.80	-15.01	-553.42	-998.98
Applied to removed ratio	1.36	0.00	0.00	0.34

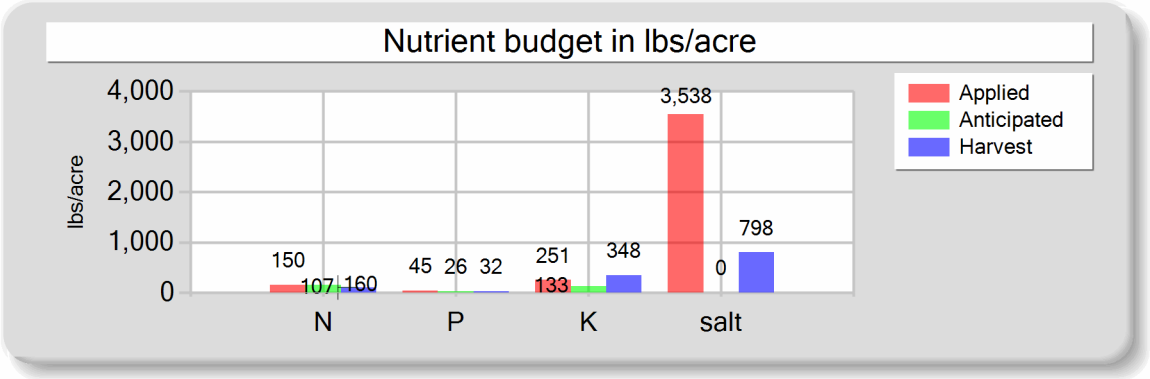
Fresh water applied	
29,240,400.00	*gallons*
1,076.82	*acre-inches*
51.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 F - 10/20/2022: Oats, silage-soft dough

Field name: 4 F | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.47	45.06	251.47	3,538.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.47	45.06	251.47	3,538.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	106.81	31.60	347.60	797.84
Nutrient balance	43.66	13.46	-96.13	2,740.50
Applied to removed ratio	1.41	1.43	0.72	4.43

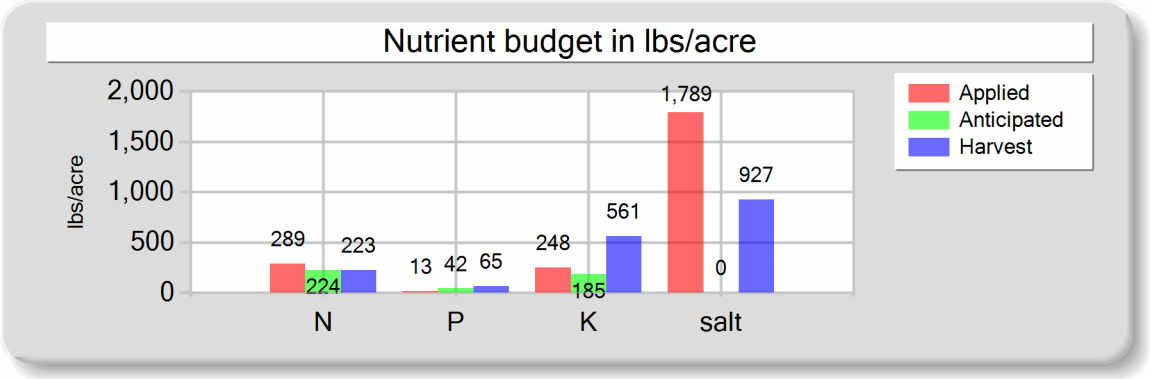
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

4 F - 06/10/2023: Corn, silage

Field name: 4 F | Crop: Corn, silage | Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	267.15	13.43	248.43	1,424.12
Fresh water	7.09	0.00	0.00	365.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.24	13.43	248.43	1,789.28
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	223.18	64.77	560.53	927.11
Nutrient balance	66.06	-51.35	-312.10	862.17
Applied to removed ratio	1.30	0.21	0.44	1.93

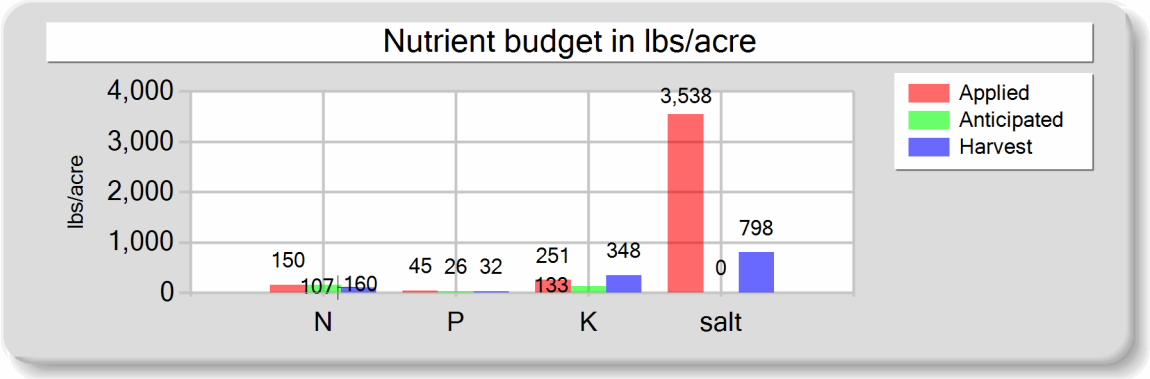
Fresh water applied	
25,245,000.00	*gallons*
929.69	*acre-inches*
30.99	*inches/acre*

Process wastewater applied	
852,000.00	*gallons*
31.38	*acre-inches*
1.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 G - 10/20/2022: Oats, silage-soft dough

Field name: 5 G | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.47	45.06	251.47	3,538.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.47	45.06	251.47	3,538.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	106.81	31.60	347.60	797.84
Nutrient balance	43.66	13.46	-96.13	2,740.50
Applied to removed ratio	1.41	1.43	0.72	4.43

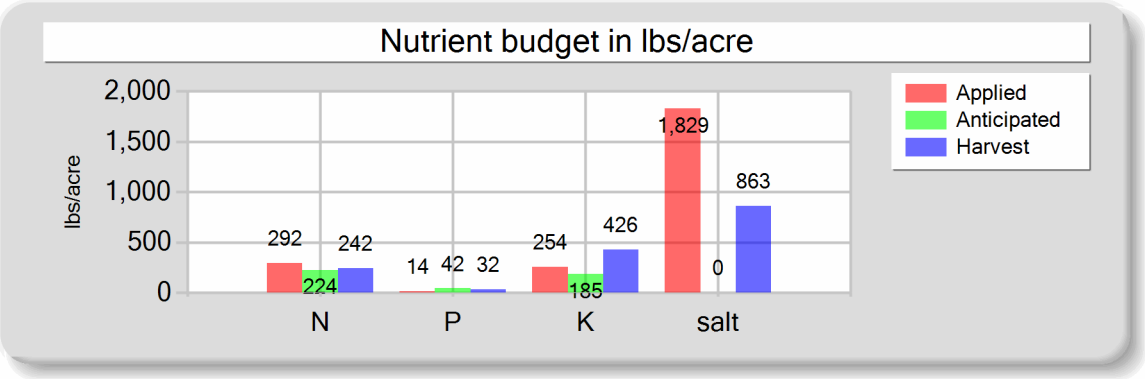
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

5 G - 06/11/2023: Corn, silage

Field name: 5 G | Crop: Corn, silage | Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	270.25	13.70	253.61	1,462.30
Fresh water	7.12	0.00	0.00	366.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.37	13.70	253.61	1,828.75
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	242.49	32.09	426.36	863.34
Nutrient balance	49.88	-18.39	-172.75	965.42
Applied to removed ratio	1.21	0.43	0.59	2.12

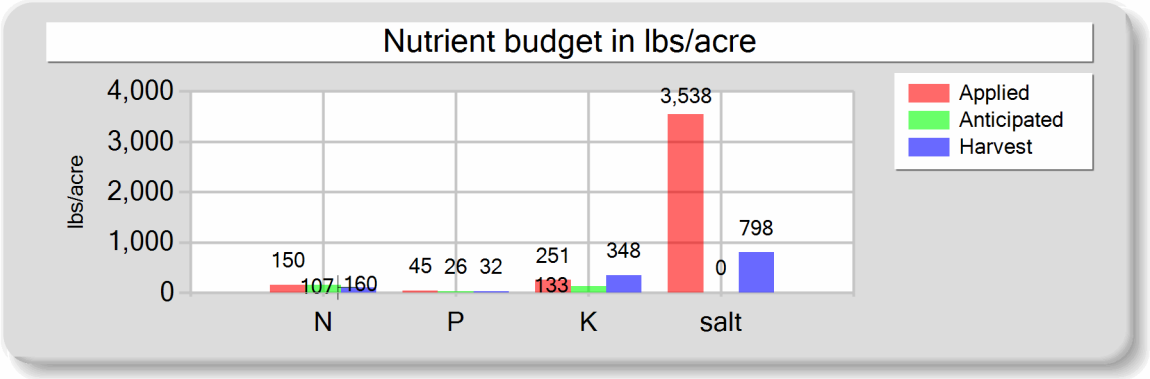
Fresh water applied	
39,690,000.00	*gallons*
1,461.65	*acre-inches*
31.10	*inches/acre*

Process wastewater applied	
1,364,500.00	*gallons*
50.25	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 H - 10/20/2022: Oats, silage-soft dough

Field name: 6 H | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.47	45.06	251.47	3,538.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.47	45.06	251.47	3,538.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	106.81	31.60	347.60	797.84
Nutrient balance	43.66	13.46	-96.13	2,740.50
Applied to removed ratio	1.41	1.43	0.72	4.43

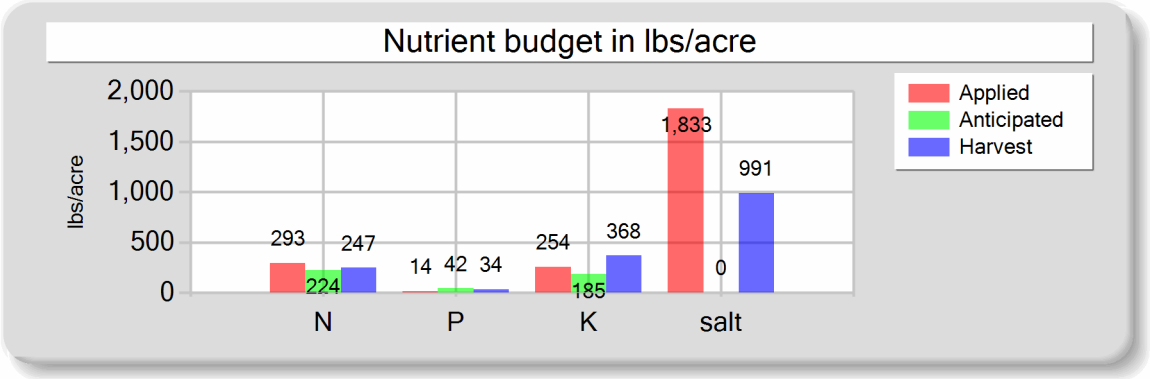
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6 H - 06/13/2023: Corn, silage

Field name: 6 H | Crop: Corn, silage | Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	270.46	13.72	253.96	1,464.91
Fresh water	7.14	0.00	0.00	367.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.60	13.72	253.96	1,832.51
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	246.88	33.75	368.13	991.25
Nutrient balance	45.73	-20.03	-114.16	841.26
Applied to removed ratio	1.19	0.41	0.69	1.85

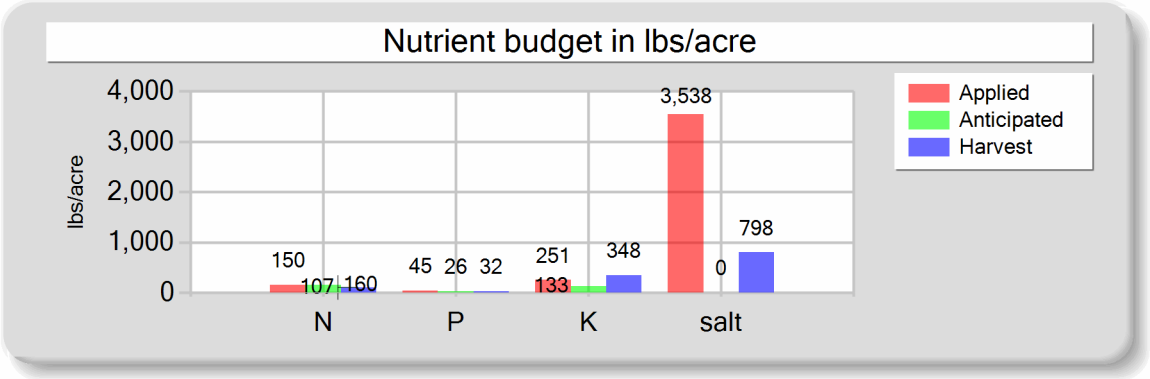
Fresh water applied	
16,942,500.00	*gallons*
623.93	*acre-inches*
31.20	*inches/acre*

Process wastewater applied	
581,500.00	*gallons*
21.41	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 I - 10/20/2022: Oats, silage-soft dough

Field name: 7 I Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.47	45.06	251.47	3,538.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.47	45.06	251.47	3,538.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	106.81	31.60	347.60	797.84
Nutrient balance	43.66	13.46	-96.13	2,740.50
Applied to removed ratio	1.41	1.43	0.72	4.43

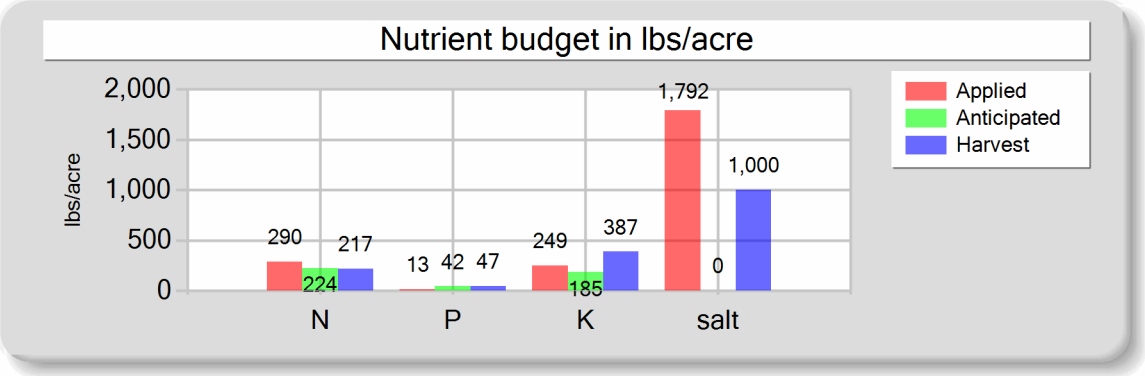
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

7 I - 06/14/2023: Corn, silage

Field name: 7 I Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	267.75	13.48	249.43	1,431.54
Fresh water	7.01	0.00	0.00	360.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.76	13.48	249.43	1,792.35
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	217.20	46.54	386.51	999.68
Nutrient balance	72.56	-33.06	-137.08	792.67
Applied to removed ratio	1.33	0.29	0.65	1.79

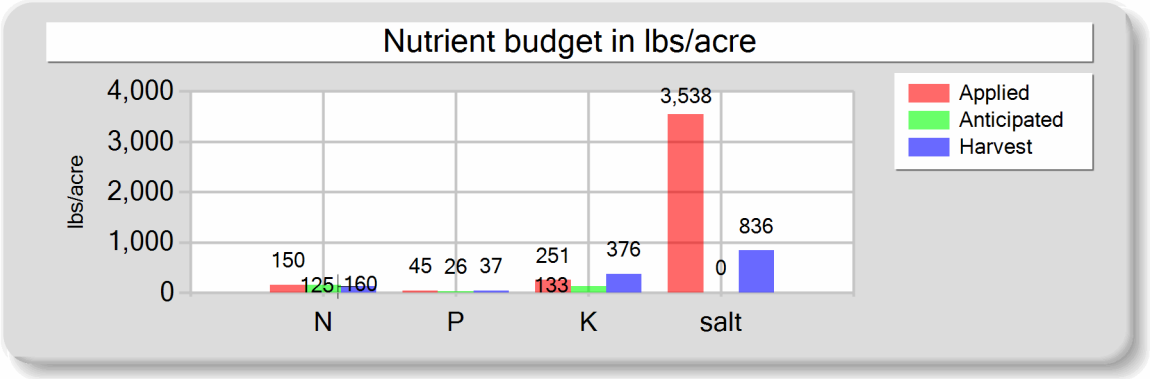
Fresh water applied	
9,146,250.00	*gallons*
336.83	*acre-inches*
30.62	*inches/acre*

Process wastewater applied	
313,750.00	*gallons*
11.55	*acre-inches*
1.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 J - 10/20/2022: Oats, silage-soft dough

Field name: 8 J Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.47	45.06	251.47	3,538.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.47	45.06	251.47	3,538.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.45	36.97	376.04	836.21
Nutrient balance	25.02	8.08	-124.57	2,702.13
Applied to removed ratio	1.20	1.22	0.67	4.23

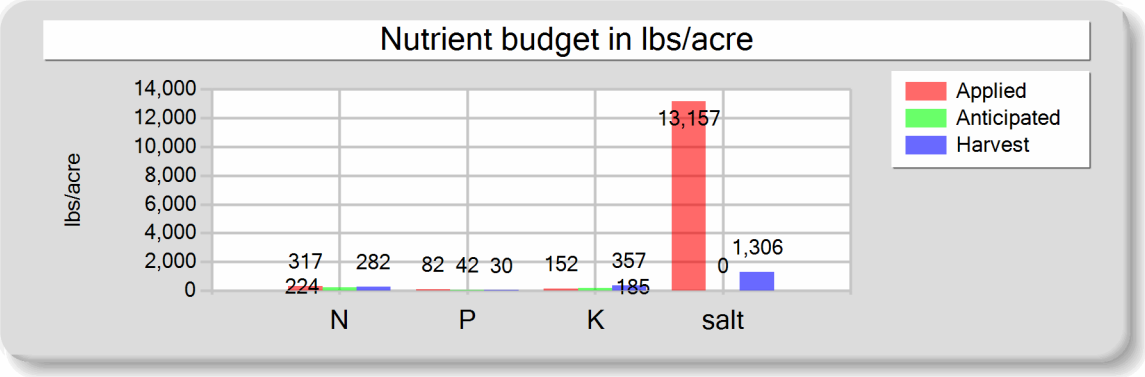
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

8 J - 06/26/2023: Corn, silage

Field name: 8 J | Crop: Corn, silage | Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	43.20	0.00	0.00	0.00
Dry manure	259.31	82.14	152.14	12,784.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.25	0.00	0.00	373.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	316.76	82.14	152.14	13,157.19
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	281.95	29.88	357.26	1,305.55
Nutrient balance	34.81	52.27	-205.12	11,851.64
Applied to removed ratio	1.12	2.75	0.43	10.08

Fresh water applied	
23,220,000.00	*gallons*
855.11	*acre-inches*
31.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	5,590.00	31,200.00							43.90
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 04/19/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	4,530.00	8,390.00							70.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,890.00	2,690.00	21,400.00							77.00
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 10/11/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	5,360.00	22,500.00							69.30
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	5,720.00	40,300.00							48.20
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/26/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	552.00	257.00	0.00	0.00	87.30	823.00								4,960.00	7,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	333.00	98.00	0.00	0.00	66.90	818.00								4,930.00	4,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	403.00	255.00	0.00	0.00	136.00	1,320.00								8,890.00	7,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 05/09/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,200.00	686.00	0.00	0.00	66.30	1,180.00								8,450.00	3,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	588.00	440.00	0.00	0.00	51.80	982.00								7,280.00	7,240
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	652.00	364.00	0.00	0.00	57.00	642.00								6.60	4,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal OID

Fresh Water

Sample description: Fresh Water

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

1 B

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

2 A

2 A

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

3 E

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

4 F

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			4.00				

5 G

5 G

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			98.00				
DL			4.00				

6 H

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

7 I

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

8 J

8 J

Soils

Sample and source description: Soils

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 B - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	1,170.00	11,900.00		8.88
DL	100.00	100.00	100.00		0.01

1 B - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	720.00	7,280.00		6.40
DL	100.00	100.00	100.00		0.01

2 A - 10/31/2018: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 12/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,630.00	1,120.00	41,300.00		12.80
DL	100.00	100.00	100.00		0.01

3 E - 01/06/2017: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 12/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,630.00	1,120.00	41,300.00		12.80
DL	100.00	100.00	100.00		0.01

4 F - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	1,000.00	11,000.00		7.89
DL	100.00	100.00	100.00		0.01

4 F - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,480.00	1,010.00	8,740.00		5.20
DL	100.00	100.00	100.00		0.01

5 G - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	1,000.00	11,000.00		7.89
DL	100.00	100.00	100.00		0.01

5 G - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	520.00	6,910.00		5.30
DL	100.00	100.00	100.00		0.01

6 H - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	1,000.00	11,000.00		7.89
DL	100.00	100.00	100.00		0.01

6 H - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,950.00	540.00	5,890.00		6.10
DL	100.00	100.00	100.00		0.01

7 I - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	1,000.00	11,000.00		7.89
DL	100.00	100.00	100.00		0.01

7 I - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,220.00	690.00	5,730.00		5.70
DL	100.00	100.00	100.00		0.01

8 J - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	1,170.00	11,900.00		8.88
DL	100.00	100.00	100.00		0.01

8 J - 06/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,530.00	480.00	5,740.00		7.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

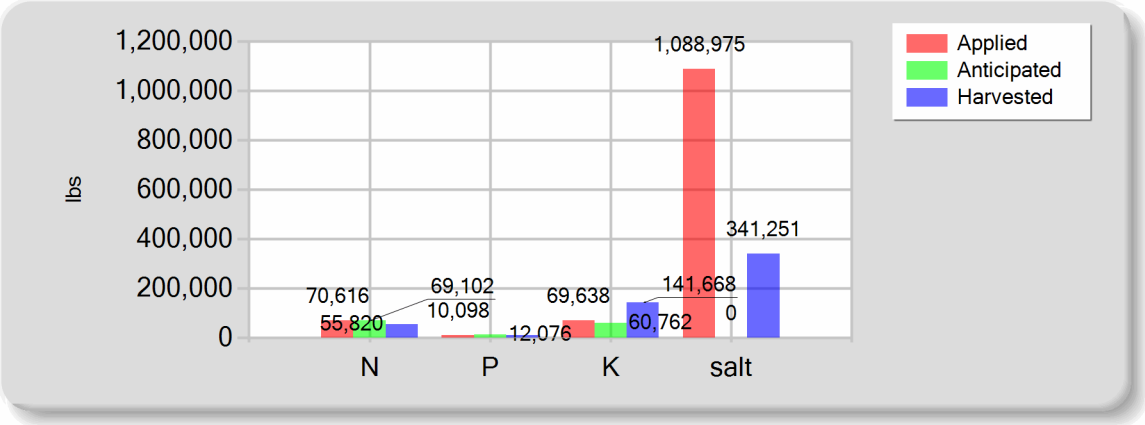
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

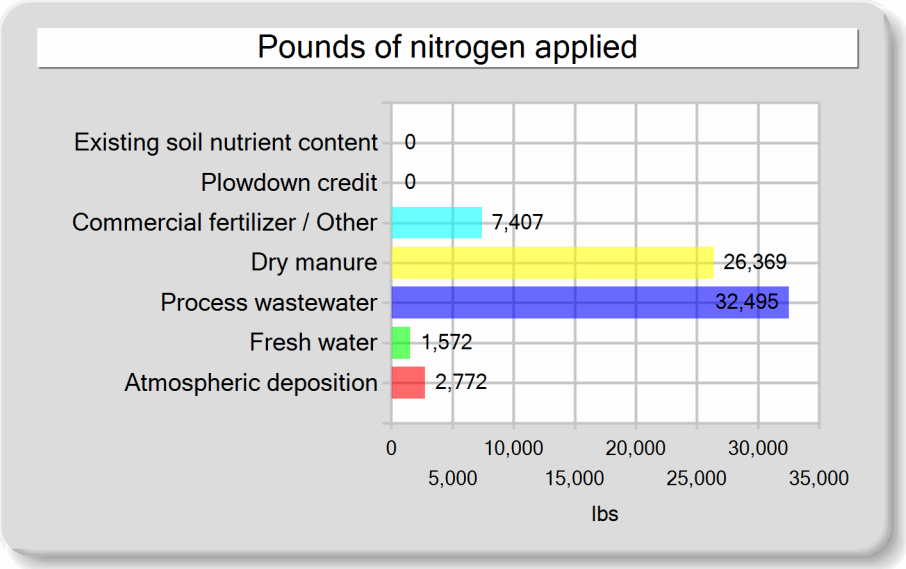
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

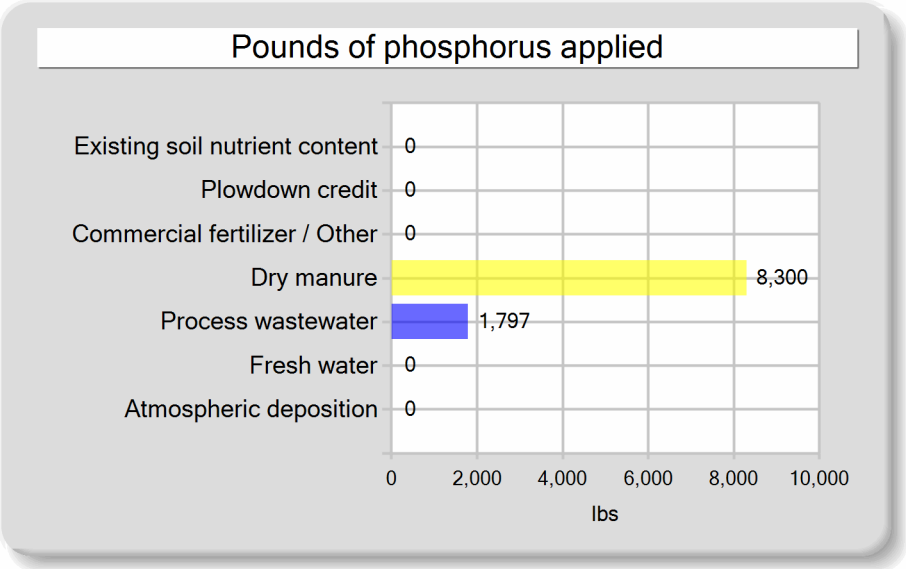
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	7,407.40	0.00	0.00	0.00
Dry manure	26,369.46	8,300.37	38,056.46	822,843.90
Process wastewater	32,494.95	1,797.34	31,581.95	185,177.55
Fresh water	1,572.36	0.00	0.00	80,953.13
Atmospheric deposition	2,772.00	0.00	0.00	0.00
Total nutrients applied	70,616.17	10,097.71	69,638.41	1,088,974.59
Anticipated crop nutrient removal	69,102.00	12,075.80	60,761.80	0.00
Actual crop nutrient removal	55,819.76	11,253.65	141,667.66	341,251.38
Nutrient balance	14,796.40	-1,155.94	-72,029.24	747,723.20
Applied to removed ratio	1.27	0.90	0.49	3.19

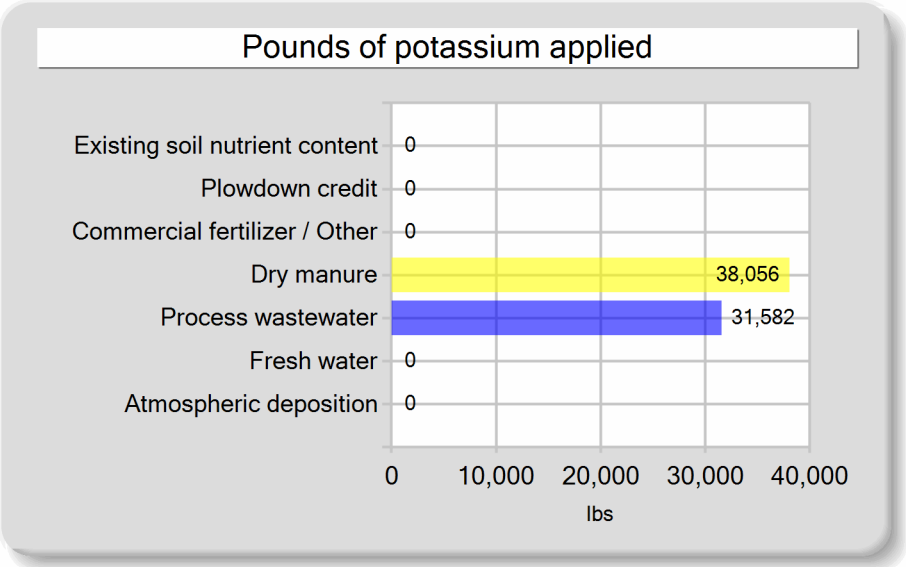
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

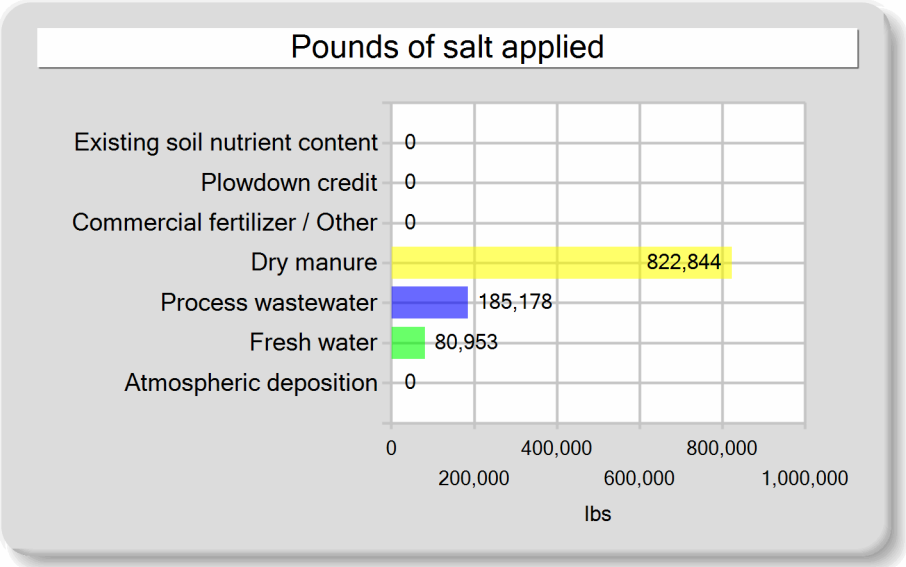


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 4F oats (1.41), Field 5G oats (1.41), Field 6H oats (1.41) and Field 7I oats (1.41) all had slightly high nitrogen field ratios as Fields failed to meet anticipated nitrogen removal of 160 pounds/acre. The subject fields removed only 106.8 pounds of nitrogen/acre. Yields for the the subject fields were slightly below anticipated levels (15.8 tons/acre compared to anticipated 16 tons/acre). The biggest factor in the elevated nitrogen field ratio was a low nitrogen value from the plant tissue lab analysis which resulted in 6.76 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton. Nitrogen application to the subject fields in the form of solid manure was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

Milk and dry cows were removed from the facility on 11/30/23 .

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY Gary Osmundson (May 16, 2024 16:49 PDT)
Vierra Brothers Dairy	Gary Osmundson
PRINT OR TYPE NAME	PRINT OR TYPE NAME
5-16-23	May 16, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Gary Osmundson

Name of Dairy Facility: 6-X Dairy #2

Facility Address: 6960 Crane Road, (Number and Street) Oakdale, CA (City) 95361 (Zip Code)

Contact Person Name and Phone Number: Gary Osmundson (Name) 209-595-6858 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Hinds Spreading

Address of Hauling Company /Person: 19075 Third Ave., (Number and Street) Stevinson, CA (City) 95374 (Zip Code)

Contact Person: Gib Hinds (Name) 209-606-7463 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Hinds Spreading (Name) 19075 Third Ave., (Number and Street) Stevinson, CA (City) 95374 (Zip Code) 209-606-7463 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

19075 Third Ave., (Number and Street) Stevinson, CA (City) 95374 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 6/5/23 to 10/31/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 8970 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 36.7% moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

Haulers Calculations

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Gary Osmundson (May 16, 2024 16:49 PDT) Date: May 16, 2024

Hauler's Signature: Gib Hinds Date: 4/20/2024

04/30/2024 2023 MANURE STATEMENT

Hinds Spreading

19075 Third Ave

Stevinson Ca 95374

209-606-7463

Osmundson Gary

6624 Crane Rd

Oakdale Ca 95361

17940 YARDS OF MANURE EXPORTED FROM DAIRY #2 TO HINDS SPREADING

2000 YARDS OF MANURE EXPORTED FROM PATTERSON RD DAIRY TO HINDS SPREADING

8000 YARDSOF MANURE EXPORTED FROM CRANE VILLA TO HINDS SPREADING

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: 6X2
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R Ag Services
Sampled By: Jeanette
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/22/23		Dairy	Domestic	24.2		✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jeanette Sanger	F&R		
2nd		DENELE ANALYTICAL, INC.	8-22-23 1:30	

(LABORATORY USE ONLY)

Logged In By:	LABORATORY NUMBER
	T3234037

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
6-X Dairy #2
P.O. Box 12290
Oakdale, CA 95361

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/22/23 1:30 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 24.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy	T3234037A	Electrical Conductivity (EC)	EPA 120.1 and	0.458	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.55	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	293	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.2 (T3234028A)	95.2 (T3234028A)	0.000
Electrical Conductivity (EC)	98.4			2.53
Nitrate Nitrogen (NO3-N)	103	99.9 (W3235002A)	99.9 (W3235002A)	0.060

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

2023 Annual Report 6-X #2 Dairy

Final Audit Report 2024-05-16

Created:	2024-05-16
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAdEAogL_nMZUKz49xg1OQgXcWf6sQYI9F

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Signer osmundsongary58@gmail.com entered name at signing as Gary Osmundson
2024-05-16 - 11:49:35 PM GMT

Document e-signed by Gary Osmundson (osmundsongary58@gmail.com)
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